Class 18: Pertussis Mini Project

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Background

Pertussis (a.k.a. whooping cough) is a common lung infection caused by the bacteria B. pertussis. This can infect all ages but is most severe for those under 1 year of age.

The CDC track the number of reported cases in the US: We can "scrape" this data with the **datapasta** package.

Data Import

```
cdc <- data.frame(
  year = c(
    1922L,
    1923L,
    1924L,
    1925L,
    1926L,
    1927L,
    1928L,
    1929L,
    1930L,
```

```
    1931L,
    1932L,
    1933L,
    1934L,
    1935L,
    1936L,
    1937L,
    1938L,
    1939L,
    1940L,
    1941L,
    1942L,
    1943L,
    1944L,
    1945L,
    1946L,
    1947L,
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    1962L,
    1963L,
    1964L,
    1965L,
    1966L,
    1967L,
    1968L,
    1969L,
    1970L,
    1971L,
    1972L,
```

```
    1973L,
    1974L,
    1975L,
    1976L,
    1977L,
    1978L,
    1979L,
    1980L,
    1981L,
    1982L,
    1983L,
    1984L,
    1985L,
    1986L,
    1987L,
    1988L,
    1989L,
    1990L,
    1991L,
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    1999L,
    2000L,
    2001L,
    2002L,
    2003L,
    2004L,
    2005L,
    2006L,
    2007L,
    2008L,
    2009L,
    2010L,
    2011L,
    2012L,
    2013L,
    2014L,
```

```r
    2015L,
    2016L,
    2017L,
    2018L,
    2019L,
    2020L,
    2021L,
    2022L,
    2023L,
    2024L,
    2025L
  ),
  cases = c(
    107473,
    164191,
    165418,
    152003,
    202210,
    181411,
    161799,
    197371,
    166914,
    172559,
    215343,
    179135,
    265269,
    180518,
    147237,
    214652,
    227319,
    103188,
    183866,
    222202,
    191383,
    191890,
    109873,
    133792,
    109860,
    156517,
    74715,
    69479,
    120718,
```

```
    68687,
    45030,
    37129,
    60886,
    62786,
    31732,
    28295,
    32148,
    40005,
    14809,
    11468,
    17749,
    17135,
    13005,
    6799,
    7717,
    9718,
    4810,
    3285,
    4249,
    3036,
    3287,
    1759,
    2402,
    1738,
    1010,
    2177,
    2063,
    1623,
    1730,
    1248,
    1895,
    2463,
    2276,
    3589,
    4195,
    2823,
    3450,
    4157,
    4570,
    2719,
    4083,
```

```r
    6586,
    4617,
    5137,
    7796,
    6564,
    7405,
    7298,
    7867,
    7580,
    9771,
    11647,
    25827,
    25616,
    15632,
    10454,
    13278,
    16858,
    27550,
    18719,
    48277,
    28639,
    32971,
    20762,
    17972,
    18975,
    15609,
    18617,
    6124,
    2116,
    3044,
    7063,
    22538,
    21996
  )
)
```

```r
head(cdc)
```

```
  year  cases
1 1922 107473
2 1923 164191
3 1924 165418
```

```
4 1925 152003
5 1926 202210
6 1927 181411
```

```r
tail(cdc)
```

```
    year cases
99  2020  6124
100 2021  2116
101 2022  3044
102 2023  7063
103 2024 22538
104 2025 21996
```

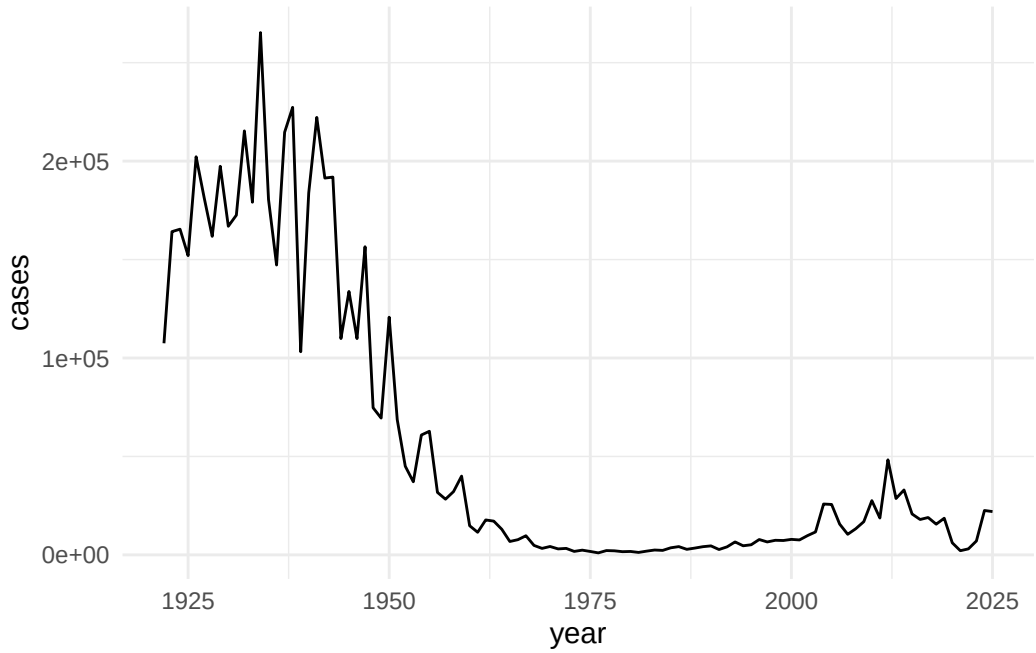
Initial Data Analysis

Q1. Make a plot of `year` vs `cases`

```r
library(ggplot2)
```

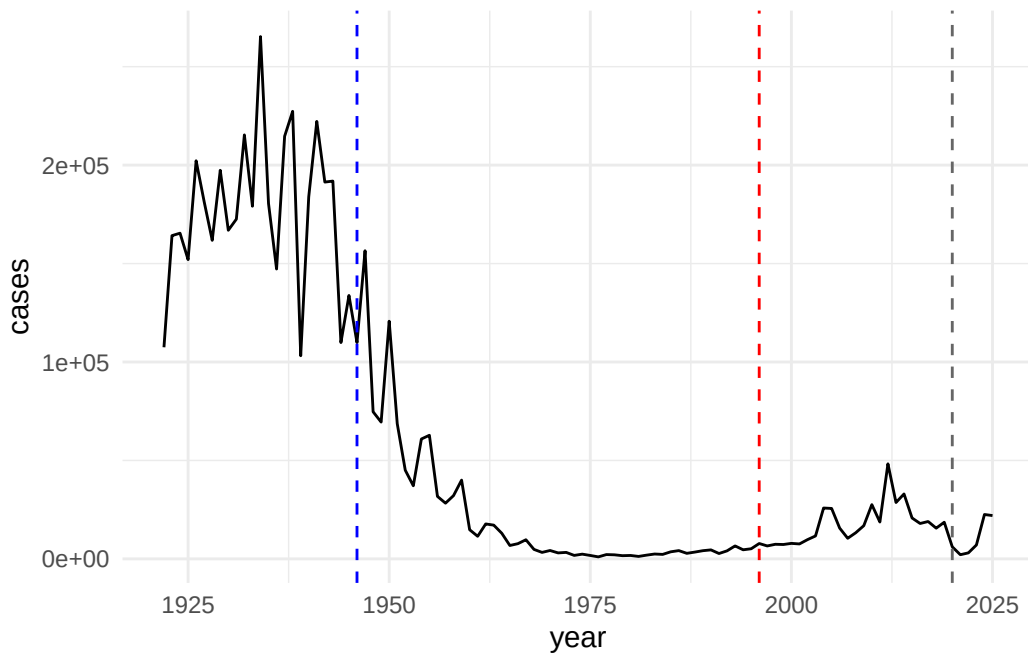
```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```r
p1 <- ggplot(cdc, aes(year, cases)) +
  geom_line() +
  theme_minimal()
p1
```



Q2. Add some major milestones including the first wP vaccine roll-out (1946), the switch to the newer aP vaccine (1996), and COVID years (2020)

```
p2 <- p1 +
  geom_vline(xintercept = 1946, col = "blue", lty = 2) +
  geom_vline(xintercept = 1996, col = "red", lty = 2) +
  geom_vline(xintercept = 2020, col = "grey40", lty = 2)
p2
```



There were high case numbers in the pre-1940s, numbers reduced dramatically after the introduction of wP, numbers started to increase after aP, and numbers dropped after the COVID-19 pandemic. It seems like protection from the aP vaccine doesn't last as long as the wP vaccine.

Why is this vaccine-preventable disease on the upswing? To answer this question, we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated indivicuals.

CMI-PB project

Computational Models of Immunity - Pertussis Boost project aims to provide the scientific community with this very information.

They make their data available via AJOON format returning API. We can read this in R with the `read_json` function from the **jsonlite** package:

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/subject",
                     simplifyVector = T)

head(subject, 3)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
```

Q3. How many "wP" and "aP" individuals are in this `subject` table?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q4. What is the biological sex breakdown?

```
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q5. In terms of race and biological sex, is this dataset representative of the US population?

```
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Let's read some more database tables:

```
specimen <- read_json("https://cmi-pb.org/api/specimen", simplifyVector = T)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = T)
```

```
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data, we need to first "join" (merge/link) the different tables to we have all the data in one place rather than spread across different tables.

We can use the `*_join` family of functions from **dplyr** to do this:

```r
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.4.3


Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
```

```
2                            1                            1         Blood
3                            3                            3         Blood
4                            7                            7         Blood
5                           11                           14         Blood
6                           32                           30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
```

```
1                           -3                              0         Blood
2                           -3                              0         Blood
3                           -3                              0         Blood
4                           -3                              0         Blood
5                           -3                              0         Blood
6                           -3                              0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q6. What antibody isotypes are measured for these patients?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Q7. What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

```r
table(abdata$dataset)
```

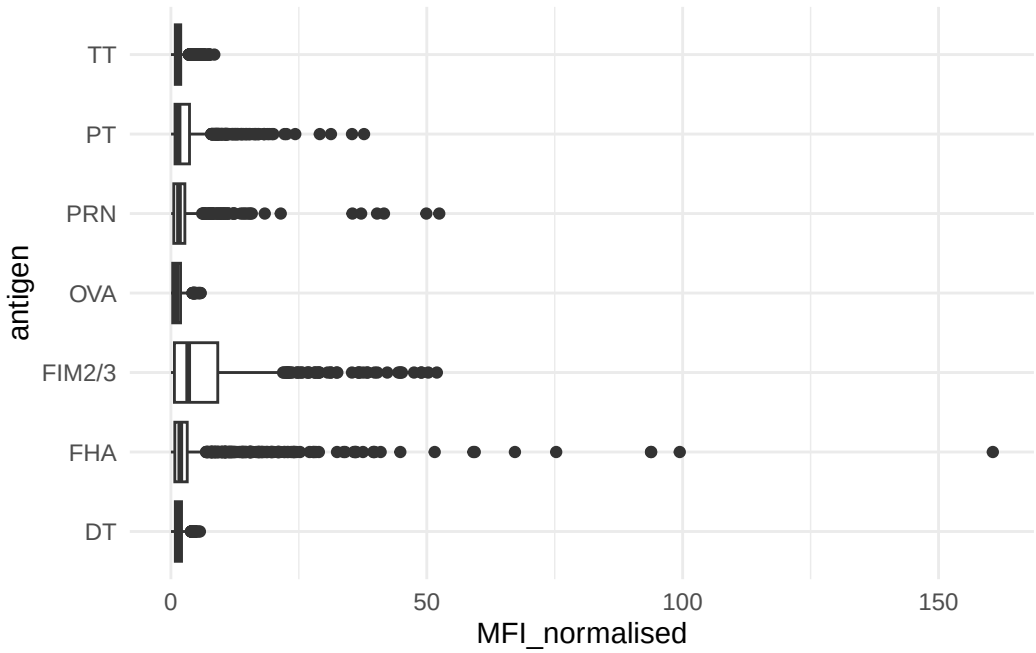
```
2020_dataset 2021_dataset 2022_dataset 2023_dataset 
       31520         8085         7301        15050 
```

It looks like there's a lot more data from 2023 than there is in 2021 and 2022, likely due to the COVID-19 pandemic.

Let's focus on the IgG isotype and make a plot of MFI_normalized for all antigens.

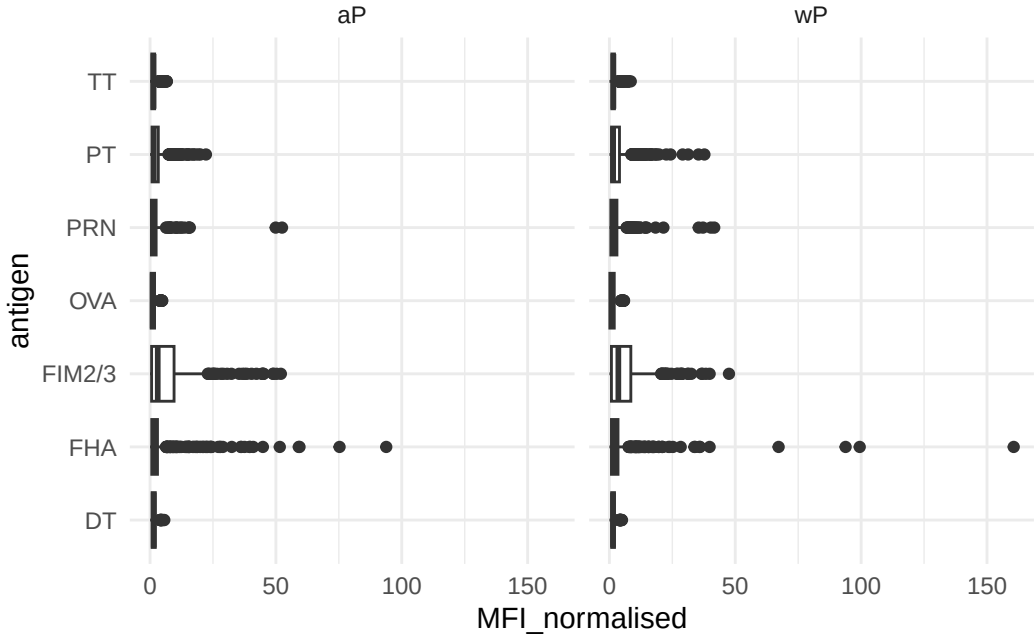
```r
igg <- abdata |>
  filter(isotype == "IgG")

p3 <- ggplot(igg, aes(MFI_normalised, antigen)) +
  geom_boxplot() +
  theme_minimal()
p3
```



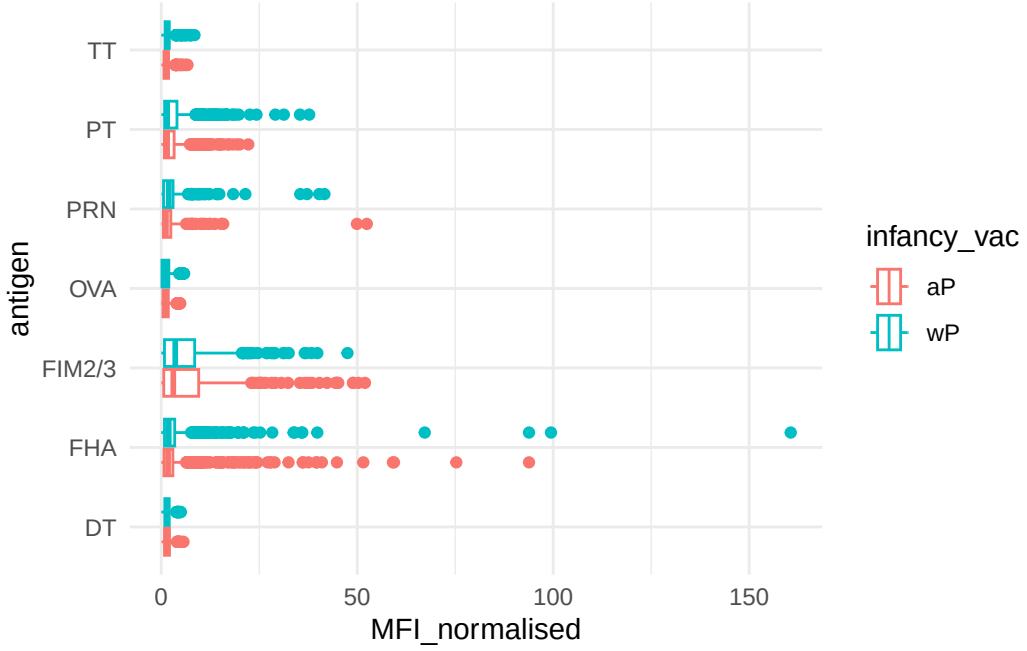
Q8. Is there a difference for aP vs wP individuals with these values?

```r
p4 <- p3 + facet_wrap(~infancy_vac)
p4
```



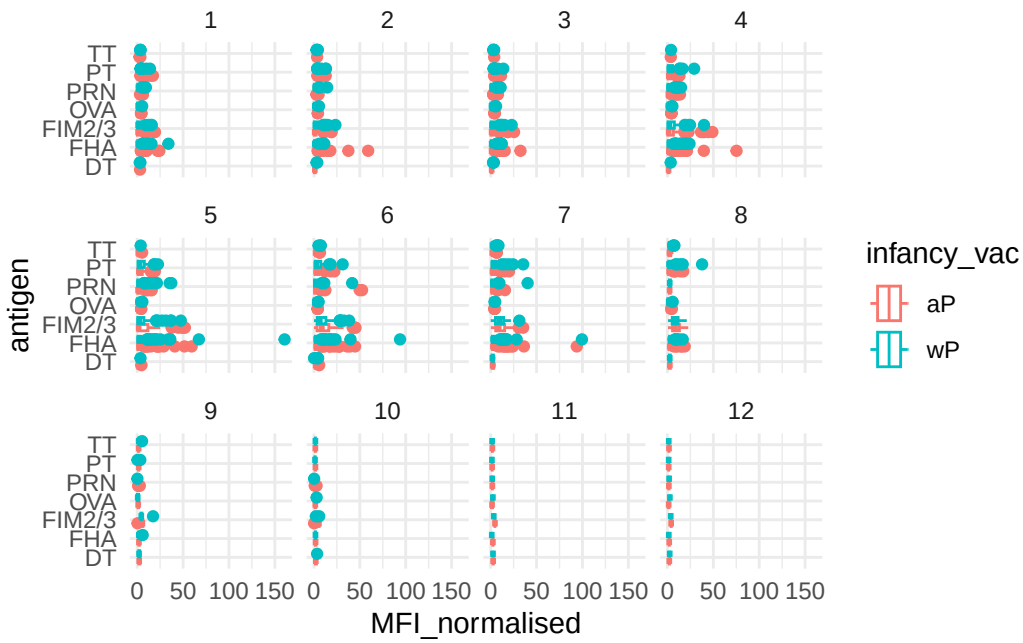
or

```
p5 <- p3 + aes(col = infancy_vac)
p5
```



Q9. Is there temporal response - i.e. do values increase or decrease over time?

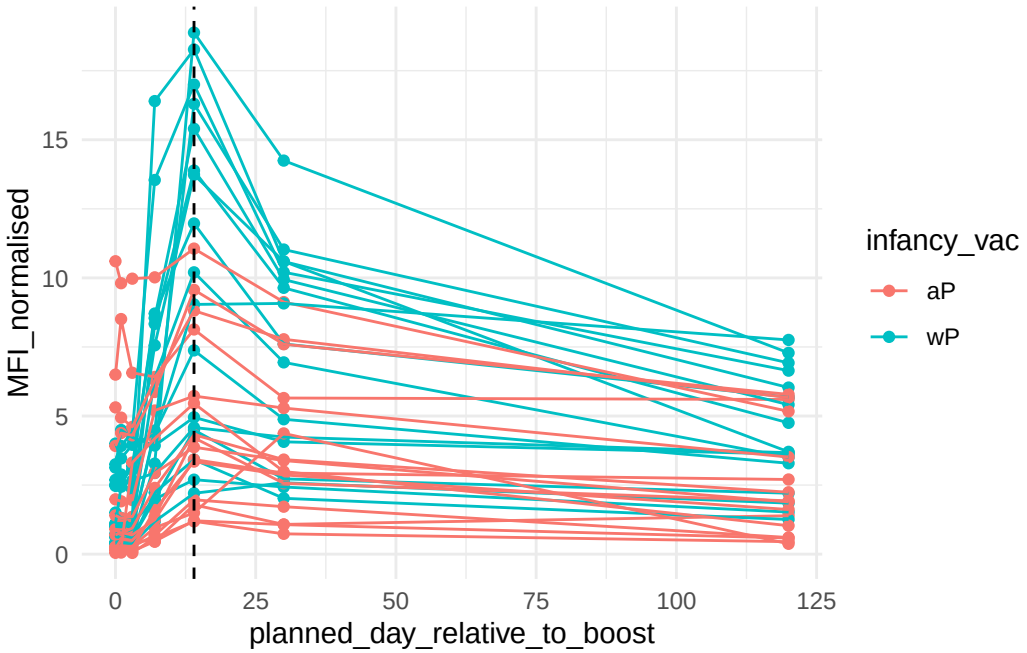
```
p6 <- p5 + facet_wrap(~visit)
p6
```



Focus on "PT" Pertussis Toxin antigen

```
pt_igg_21 <- igg |>
  filter(antigen == "PT",
         dataset == "2021_dataset")
```

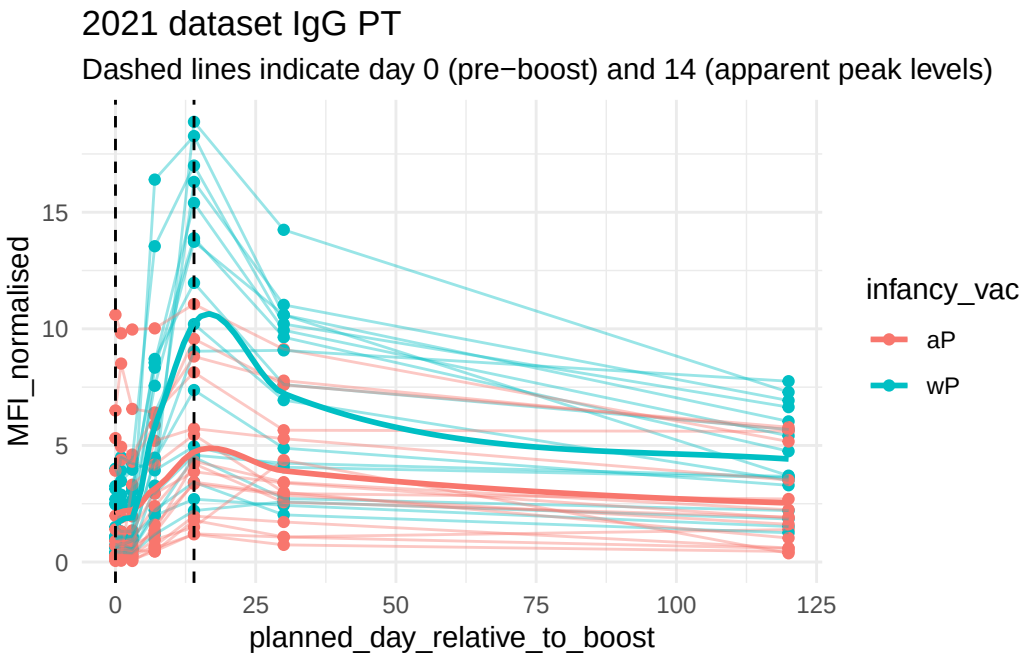
```
p7 <- ggplot(pt_igg_21, aes(planned_day_relative_to_boost, MFI_normalised, col = infancy_vac
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 14, col = "black", lty = 2) +
  theme_minimal()
p7
```




```
igg_avg <- pt_igg_21 %>%
  group_by(infancy_vac) %>%
  mutate(mean_MFI = mean(MFI_normalised))

p8 <- ggplot(igg_avg, aes(planned_day_relative_to_boost, MFI_normalised, col = infancy_vac, g
  geom_point() +
  geom_line(alpha = 0.4) +
  geom_vline(xintercept = 0, col = "black", lty = 2) +
  geom_vline(xintercept = 14, col = "black", lty = 2) +
  labs(title="2021 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)") +
  theme_minimal() +
  geom_smooth(aes(group = infancy_vac),
              se = F,
              span = .4)
p8
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


2021 dataset IgG PT
Dashed lines indicate day 0 (pre−boost) and 14 (apparent peak levels)
MFI_normalised
15
10
5
0
0
25
50
75
100
125
planned_day_relative_to_boost
infancy_vac
aP
wP